



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

summer2025-2026

Section: [AA]

Group No: 12

Project Report On

Project Name [Clinic Management System]

Supervised By

Dr.Md.Iftekharul Mobin

Submitted By:

Name			ID		
1.Md Fahimuzzaman Fahim			23-54654-3		
2. Md Sabbir Ahmed Rifat			24-56062-1		
3. ISMITA MANNAN MIM			23-51088-1		
Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of ObjectOriented Programming project development activities.

Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				

Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application	Limited understanding of graphical user interfaces.	Shows a basic understanding of creating user interfaces. Most	Effectively identifies and meet the consider the	Exhibits an exceptional work design following a high
	interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

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Chapter 1: Introduction

A clinic is responsible for managing doctors, patients, appointments, and daily administrative tasks. When these activities are handled manually, it can lead to delays, record mismatches, and difficulties in maintaining accurate information. Therefore, an organized and automated system is essential to ensure smooth clinic operations.

The Clinic Management System is developed to digitalize and automate the main functions of a clinic. The system includes doctor, receptionist, admin, and patient registration with full CRUD operations. Receptionists can register patients, create appointments with doctors, and generate bills, while users can view patient and appointment details easily.

Administrators have full control over doctor, receptionist, and admin modules, ensuring proper management and security through role-based access. By implementing this system, the clinic can reduce paperwork, improve data accuracy, and enhance service quality. Overall, the project aims to provide a user-friendly and efficient solution for effective clinic management.

Chapter2: User Story

Admin

- **An Admin can perform Add, Update, and Delete operations on system entities such as Doctors, Receptionists, and Admin accounts.**
 - **An Admin can manage user accounts (Receptionists and other Admins) with full CRUD control.**
 - **An Admin is responsible for system and user management through role-based access control.**
 - **The Admin cannot access Appointment and Billing modules.**
 - **The Admin holds the highest authority in the system for user and role management.**
-

Receptionist

- **A Receptionist comes next to the Admin with limited system access.**
- **A Receptionist can add, update, and manage Patient details, including registration and personal information.**
- **A Receptionist can create and manage Appointments between doctors and patients.**
- **A Receptionist can generate Bills for patients, including payment amount and status.**
- **Receptionists have restricted access and cannot manage Doctor, Admin, or Receptionist records, including their own profile.**

Chapter3(a):ER diagram

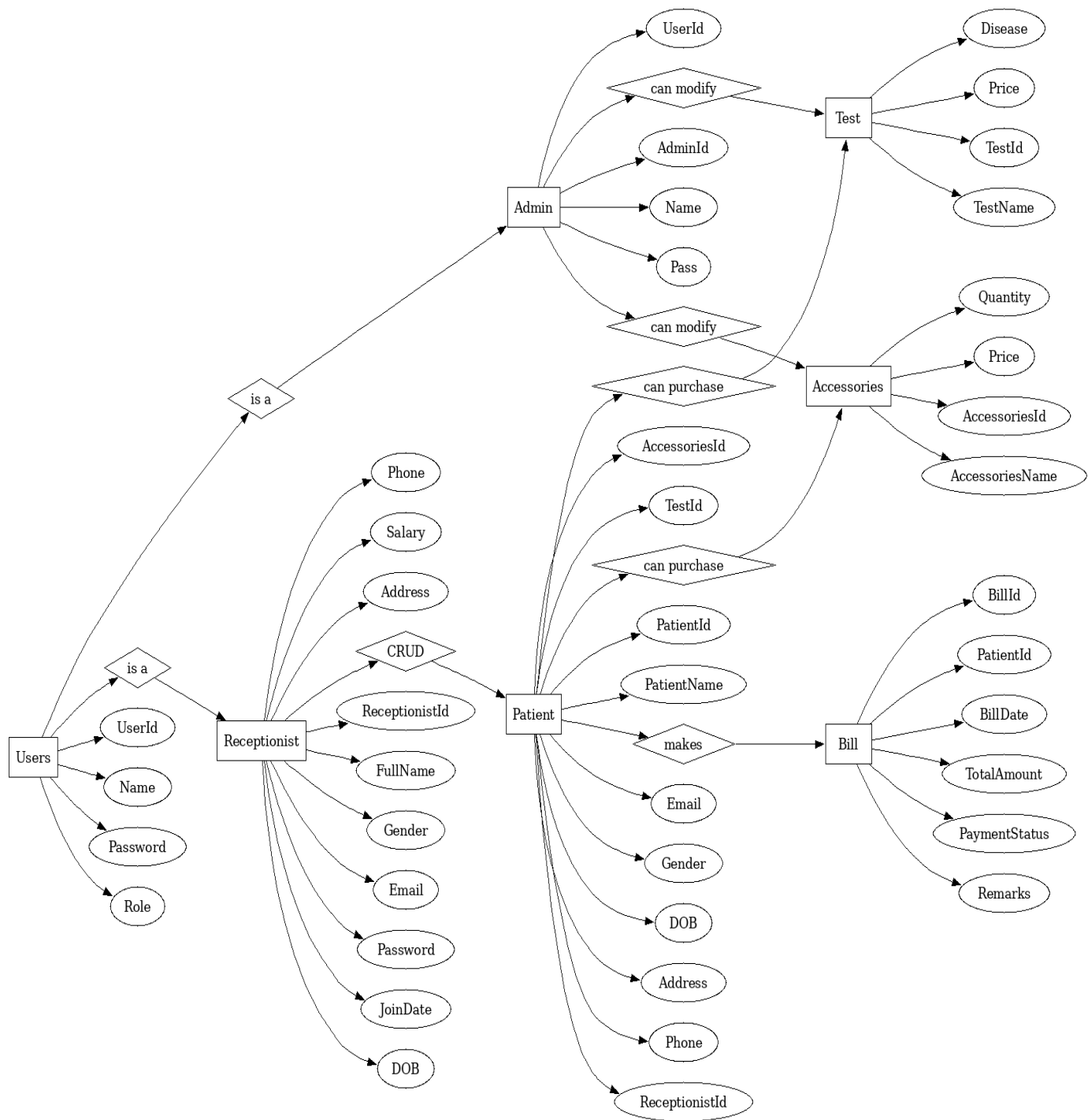


Figure 1:ER diagram

Chapter3(b):Queries

```
USE [master]
GO
/***** Object: Database [Clinic]  Script Date: 1/19/2026 9:32:43 PM *****/
CREATE DATABASE [Clinic]
CONTAINMENT = NONE
ON PRIMARY
( NAME = N'NewDignostic', FILENAME = N'C:\Program Files\Microsoft SQL
Server\MSSQL17.SQLEXPRESS\MSSQL\DATA\NewDignostic.mdf' , SIZE = 8192KB ,
MAXSIZE = UNLIMITED, FILEGROWTH = 65536KB )
LOG ON
( NAME = N'NewDignostic_log', FILENAME = N'C:\Program Files\Microsoft SQL
Server\MSSQL17.SQLEXPRESS\MSSQL\DATA\NewDignostic_log.ldf' , SIZE = 8192KB ,
MAXSIZE = 2048GB , FILEGROWTH = 65536KB )
WITH CATALOG_COLLATION = DATABASE_DEFAULT, LEDGER = OFF
GO
ALTER DATABASE [Clinic] SET COMPATIBILITY_LEVEL = 160
GO
IF (1 = FULLTEXTSERVICEPROPERTY('IsFullTextInstalled'))
begin
EXEC [Clinic].[dbo].[sp_fulltext_database] @action = 'enable'
end
GO
ALTER DATABASE [Clinic] SET ANSI_NULL_DEFAULT OFF
GO
ALTER DATABASE [Clinic] SET ANSI_NULLS OFF
GO
ALTER DATABASE [Clinic] SET ANSI_PADDING OFF
GO
ALTER DATABASE [Clinic] SET ANSI_WARNINGS OFF
GO
ALTER DATABASE [Clinic] SET ARITHABORT OFF
GO
```

```
ALTER DATABASE [Clinic] SET AUTO_CLOSE OFF
GO
ALTER DATABASE [Clinic] SET AUTO_SHRINK OFF
GO
ALTER DATABASE [Clinic] SET AUTO_UPDATE_STATISTICS ON
GO
ALTER DATABASE [Clinic] SET CURSOR_CLOSE_ON_COMMIT OFF
GO
ALTER DATABASE [Clinic] SET CURSOR_DEFAULT GLOBAL
GO
ALTER DATABASE [Clinic] SET CONCAT_NULL_YIELDS_NULL OFF
GO
ALTER DATABASE [Clinic] SET NUMERIC_ROUNDABORT OFF
GO
ALTER DATABASE [Clinic] SET QUOTED_IDENTIFIER OFF
GO
ALTER DATABASE [Clinic] SET RECURSIVE_TRIGGERS OFF
GO
ALTER DATABASE [Clinic] SET DISABLE_BROKER
GO
ALTER DATABASE [Clinic] SET AUTO_UPDATE_STATISTICS_ASYNC OFF
GO
ALTER DATABASE [Clinic] SET DATE_CORRELATION_OPTIMIZATION OFF
GO
ALTER DATABASE [Clinic] SET TRUSTWORTHY OFF
GO
ALTER DATABASE [Clinic] SET ALLOW_SNAPSHOT_ISOLATION OFF
GO
ALTER DATABASE [Clinic] SET PARAMETERIZATION SIMPLE
GO
ALTER DATABASE [Clinic] SET READ_COMMITTED_SNAPSHOT OFF
GO
ALTER DATABASE [Clinic] SET HONOR_BROKER_PRIORITY OFF
GO
ALTER DATABASE [Clinic] SET RECOVERY SIMPLE
GO
ALTER DATABASE [Clinic] SET MULTI_USER
GO
ALTER DATABASE [Clinic] SET PAGE_VERIFY CHECKSUM
GO
ALTER DATABASE [Clinic] SET DB_CHAINING OFF
```

```

GO
ALTER DATABASE [Clinic] SET FILESTREAM( NON_TRANSACTED_ACCESS = OFF )
GO
ALTER DATABASE [Clinic] SET TARGET_RECOVERY_TIME = 60 SECONDS
GO
ALTER DATABASE [Clinic] SET DELAYED_DURABILITY = DISABLED
GO
ALTER DATABASE [Clinic] SET ACCELERATED_DATABASE_RECOVERY = OFF
GO
ALTER DATABASE [Clinic] SET OPTIMIZED_LOCKING = OFF
GO
ALTER DATABASE [Clinic] SET QUERY_STORE = ON
GO
ALTER DATABASE [Clinic] SET QUERY_STORE (OPERATION_MODE = READ_WRITE,
CLEANUP_POLICY = (STALE_QUERY_THRESHOLD_DAYS = 30),
DATA_FLUSH_INTERVAL_SECONDS = 900, INTERVAL_LENGTH_MINUTES = 60,
MAX_STORAGE_SIZE_MB = 1000, QUERY_CAPTURE_MODE = AUTO,
SIZE_BASED_CLEANUP_MODE = AUTO, MAX_PLANS_PER_QUERY = 200,
WAIT_STATS_CAPTURE_MODE = ON)
GO
USE [Clinic]
GO
/***** Object: Table [dbo].[Admin]  Script Date: 1/19/2026 9:32:43 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Admin](
    [AdminId] [varchar](20) NOT NULL,
    [Name] [varchar](100) NOT NULL,
    [Pass] [varchar](100) NOT NULL,
    [UserId] [varchar](20) NOT NULL,
    PRIMARY KEY CLUSTERED
(
    [AdminId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[Bill]  Script Date: 1/19/2026 9:32:44 PM *****/

```

```

SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Bill](
    [BillId] [varchar](20) NOT NULL,
    [PatientId] [varchar](20) NOT NULL,
    [BillDate] [date] NULL,
    [TotalAmount] [decimal](10, 2) NULL,
    [PaymentStatus] [varchar](20) NULL,
    CONSTRAINT [PK__Bill__11F2FC6A852784E5] PRIMARY KEY CLUSTERED
(
    [BillId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[DOCTOR]    Script Date: 1/19/2026 9:32:44 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[DOCTOR](
    [ID] [varchar](10) NOT NULL,
    [NAME] [varchar](50) NOT NULL,
    [SPECIALIZATION] [varchar](50) NULL,
    [FEES] [decimal](8, 2) NULL,
    PRIMARY KEY CLUSTERED
(
    [ID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[Patient]    Script Date: 1/19/2026 9:32:44 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO

```

```

CREATE TABLE [dbo].[Patient](
    [PatientId] [varchar](20) NOT NULL,
    [PatientName] [varchar](100) NOT NULL,
    [Phone] [varchar](20) NULL,
    [Email] [varchar](100) NULL,
    [Gender] [varchar](10) NULL,
    [DateOfBirth] [date] NULL,
    [Address] [varchar](255) NULL,
    [ReceptionistId] [varchar](20) NULL,
    CONSTRAINT [PK_Patient] PRIMARY KEY CLUSTERED
(
    [PatientId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[Receptionist]  Script Date: 1/19/2026 9:32:44 PM
*****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Receptionist](
    [ReceptionistId] [varchar](20) NOT NULL,
    [FullName] [varchar](100) NOT NULL,
    [Gender] [varchar](10) NULL,
    [Email] [varchar](100) NULL,
    [Pass] [varchar](100) NOT NULL,
    [JoinDate] [date] NULL,
    [DOB] [date] NULL,
    [Phone] [varchar](20) NULL,
    [Salary] [decimal](10, 2) NULL,
    [Address] [varchar](255) NULL,
    PRIMARY KEY CLUSTERED
(
    [ReceptionistId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY],
UNIQUE NONCLUSTERED

```

```

(
    [Email] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[Users]  Script Date: 1/19/2026 9:32:44 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Users](
    [UserId] [varchar](20) NOT NULL,
    [Name] [varchar](100) NOT NULL,
    [Password] [varchar](100) NOT NULL,
    [Role] [varchar](100) NOT NULL,
    CONSTRAINT [PK__Users__1788CC4CDD471C49] PRIMARY KEY CLUSTERED
(
    [UserId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
ALTER TABLE [dbo].[Bill] ADD CONSTRAINT [DF__Bill__BillDate__49C3F6B7] DEFAULT
(getdate()) FOR [BillDate]
GO
ALTER TABLE [dbo].[Bill] ADD CONSTRAINT [DF_Bill_PaymentStatus] DEFAULT ('Paid')
FOR [PaymentStatus]
GO
ALTER TABLE [dbo].[Admin] WITH CHECK ADD CONSTRAINT [FK_Admin_User]
FOREIGN KEY([UserId])
REFERENCES [dbo].[Users] ([UserId])
GO
ALTER TABLE [dbo].[Admin] CHECK CONSTRAINT [FK_Admin_User]
GO
ALTER TABLE [dbo].[Receptionist] WITH CHECK ADD CONSTRAINT
[FK_Receptionist_User] FOREIGN KEY([ReceptionistId])
REFERENCES [dbo].[Users] ([UserId])
GO

```

```
ALTER TABLE [dbo].[Receptionist] CHECK CONSTRAINT [FK_Receptionist_User]
GO
ALTER TABLE [dbo].[DOCTOR] WITH CHECK ADD CHECK ((([FEES]>=(0))))
GO
USE [master]
GO
ALTER DATABASE [Clinic] SET READ_WRITE
GO
```

Chapter 4: Screenshots

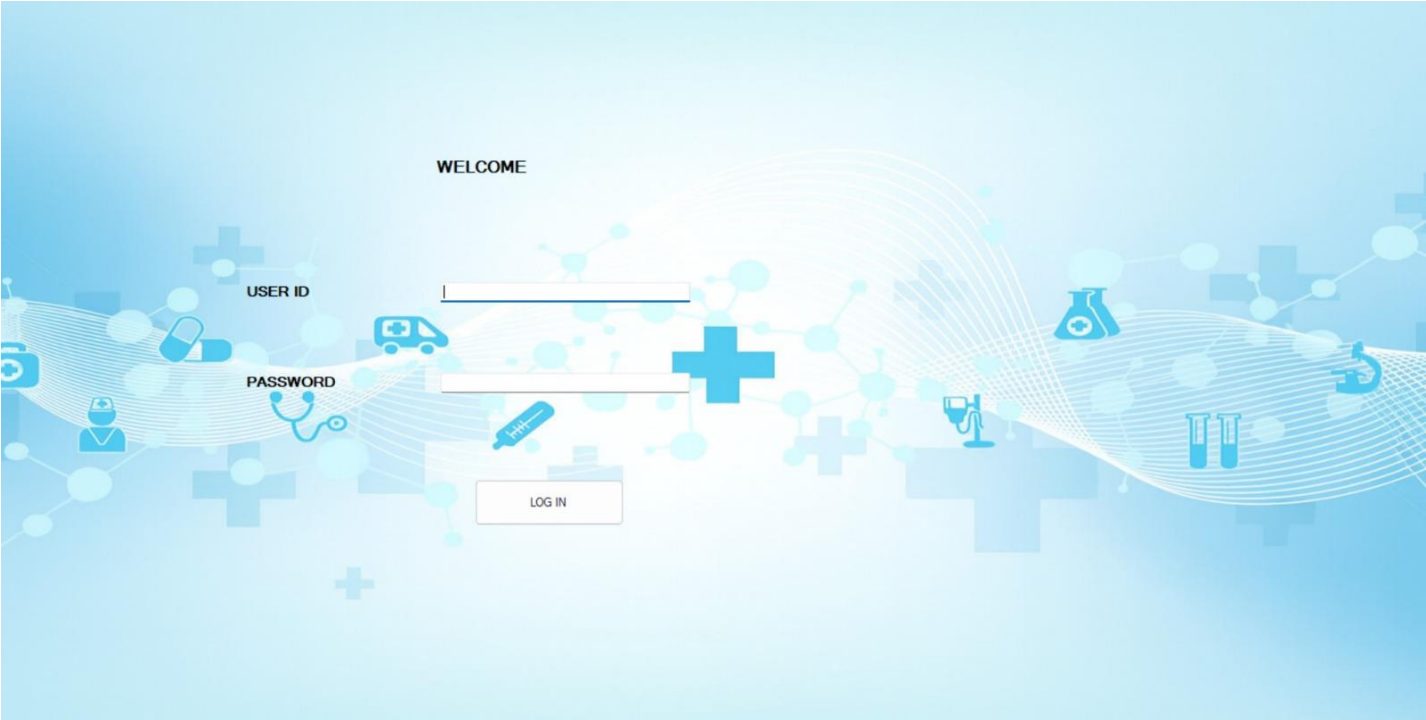


Figure1: Login Panel

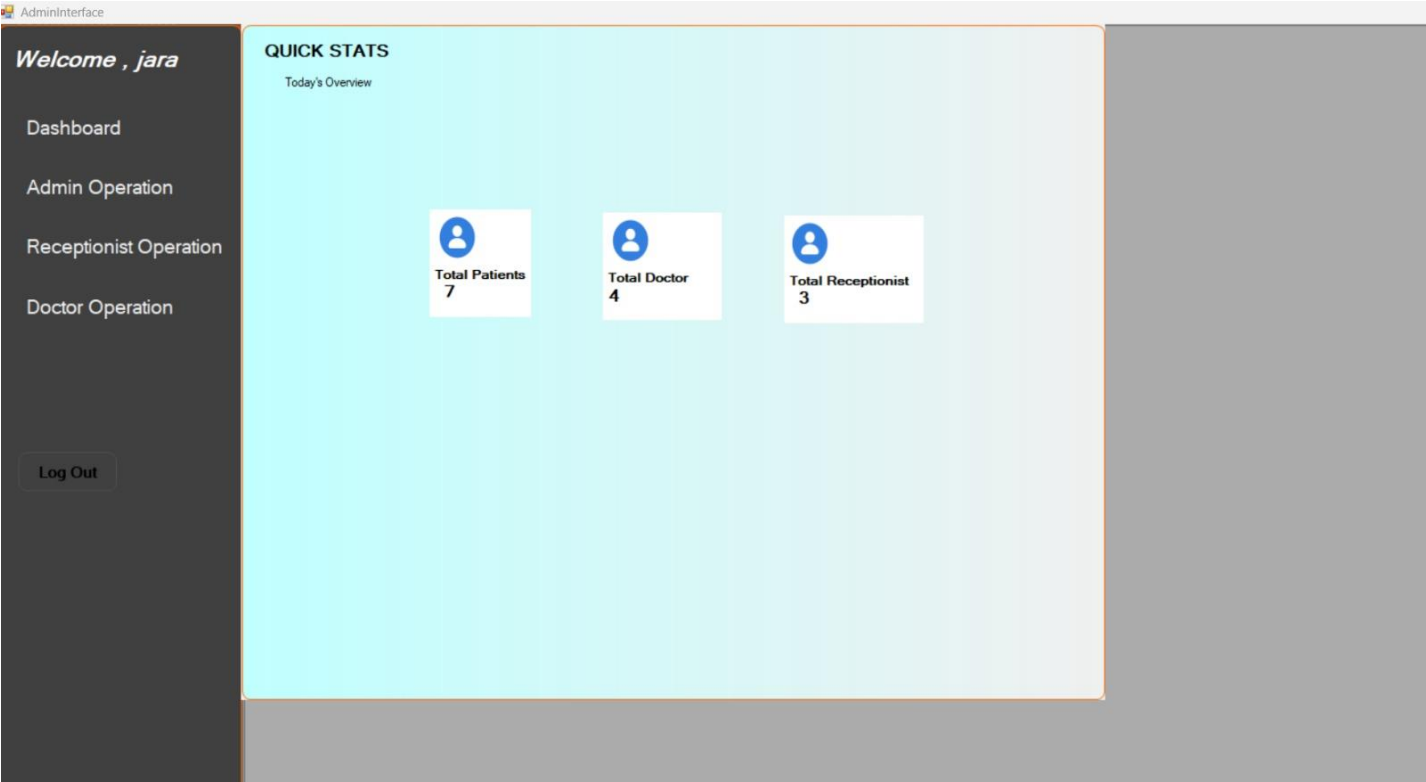


Figure 2: Admin Panel

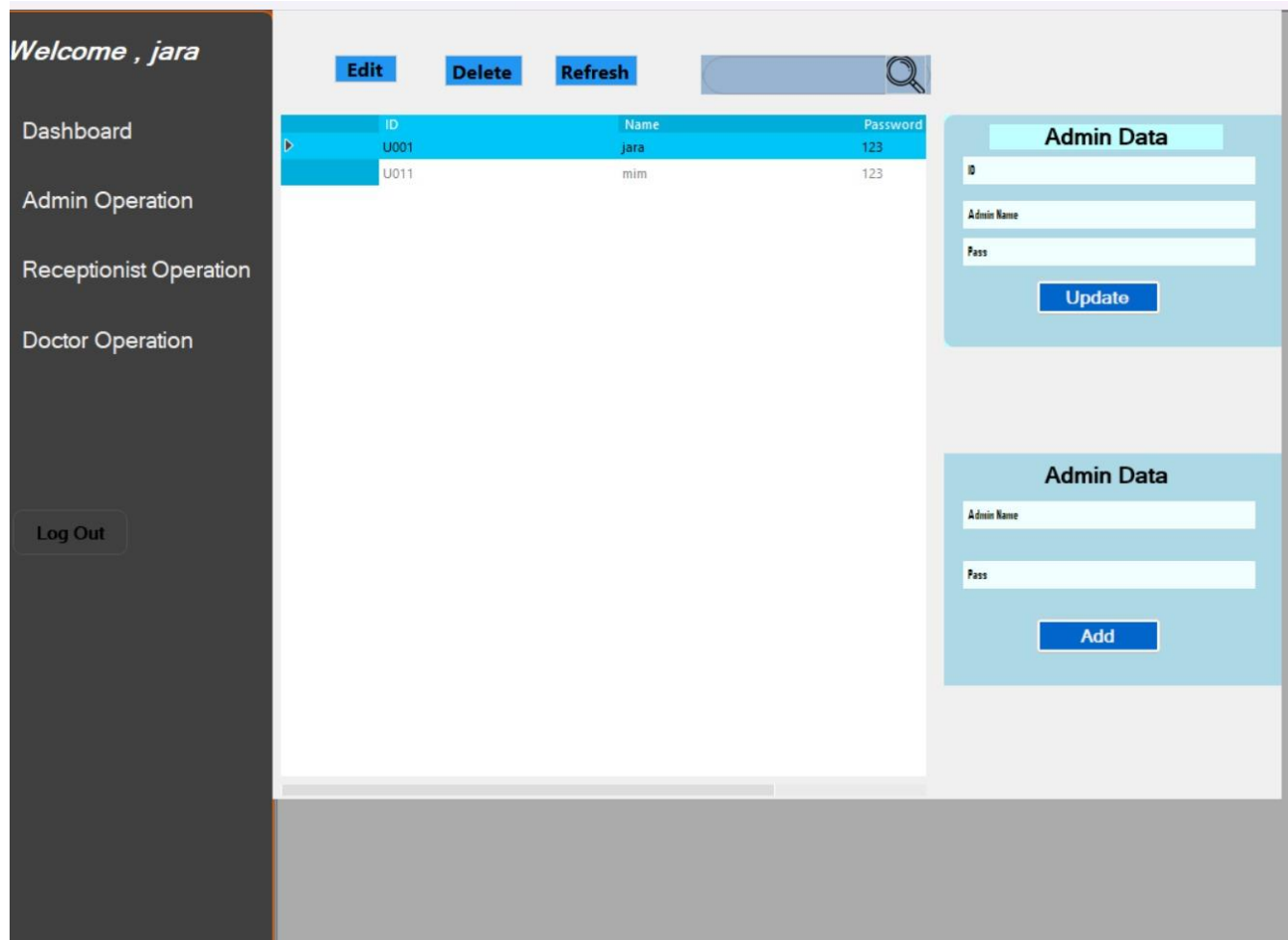


Figure 3: Admin Operation

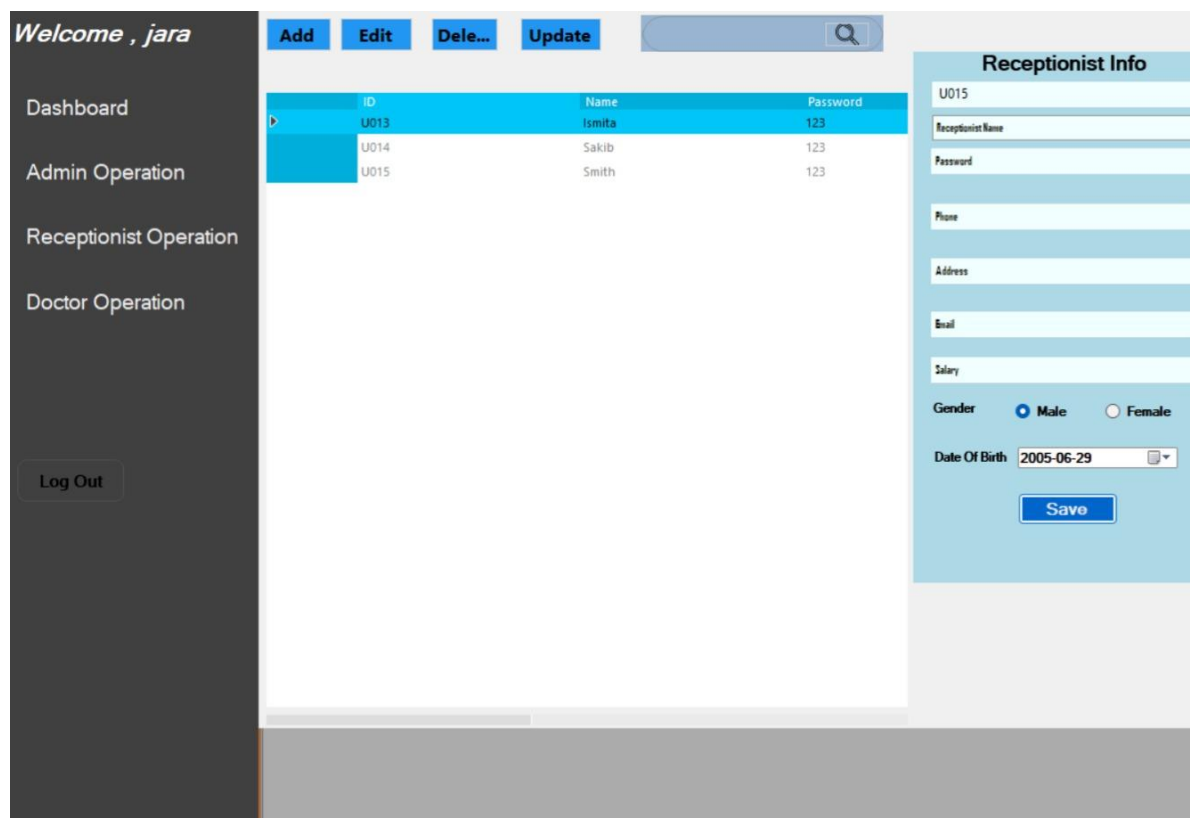


Figure 5 Receptionist Operation

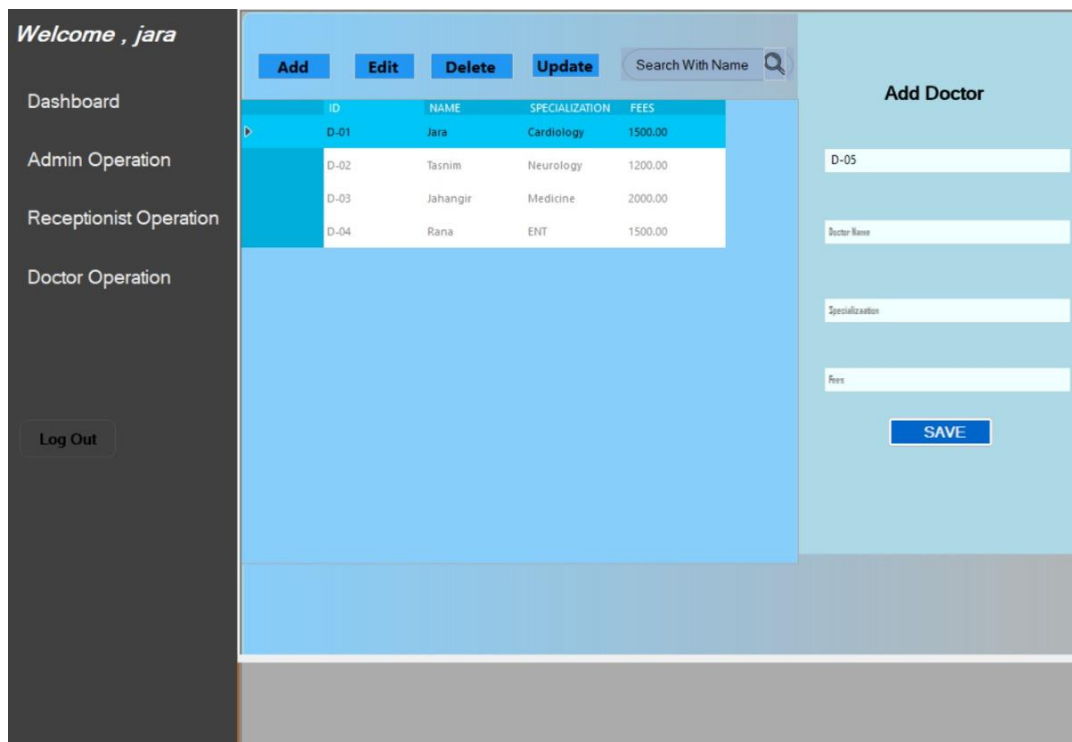
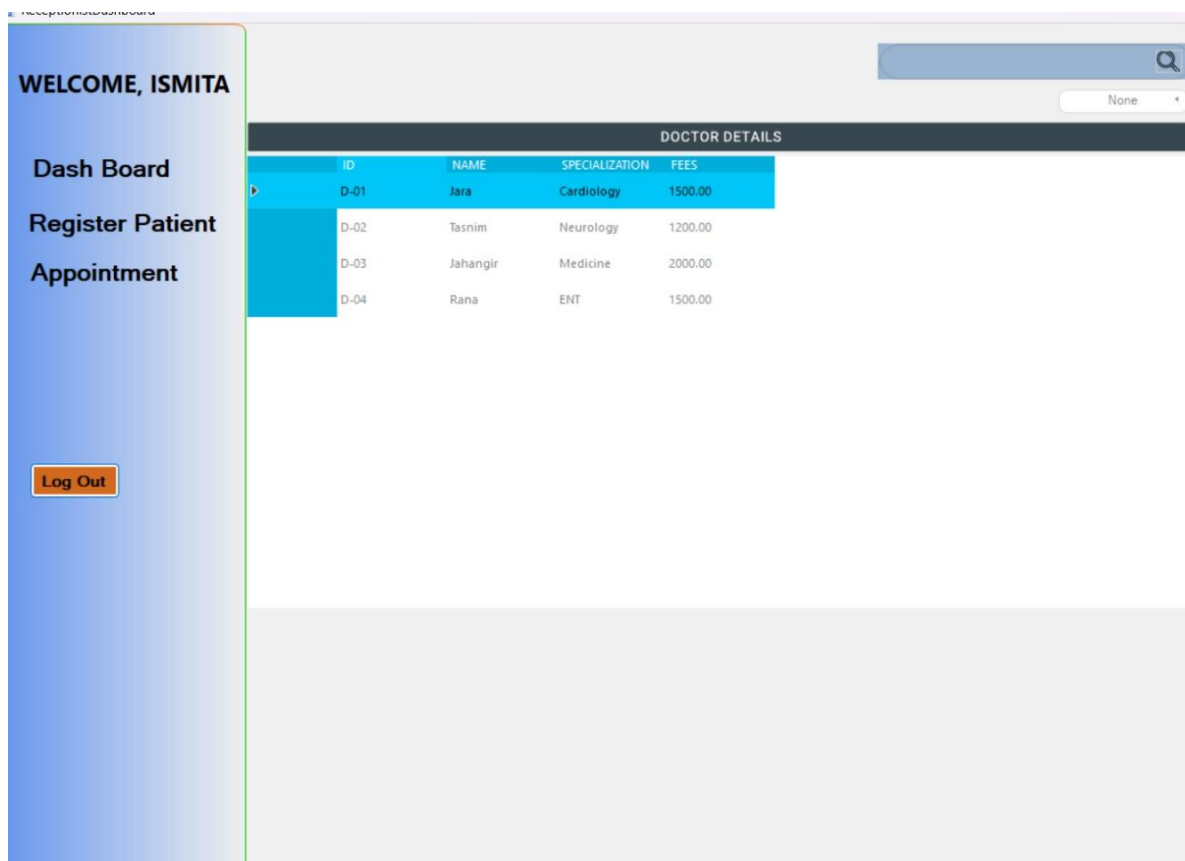


Figure 6 : Doctor operation



ReceptionistDashboard

WELCOME, ISMITA

Dash Board

Register Patient

Appointment

Log Out

ADD

Edit

Delete

Refresh

ID	Patient Name	Phone	Email
P001	Smith	019888789	smith02
P003	Sakib	1234	ds
P004	Mim	123	dd
P006	Zampa	0986531	jsm
P007	sakib	12345678	
P008	yihh	748499490	ggh
P009	sakib	456	

PATIENT FORM

Patient Name

Phone

Email

Address

Gender

☐ Male
 ☐ Female

Date Of Birth

2026-01-19

Update Info

Figure 8: Register patient

ReceptionistDashboard

WELCOME, ISMITA

Dash Board

Register Patient

Appointment

Log Out

All doctor List

ID	NAME	SPECIALIZATION	FEES
D-01	Jara	Cardiology	1500.00
D-02	Tasnim	Neurology	1200.00
D-03	Jahangir	Medicine	2000.00
D-04	Rana	ENT	1500.00

Patient Info

Existing Patient ID

For New Patient

New Patient Name

Phone

Selected Doctor

Name	Quantity	Price

Booking

Remove

Confirm

Figure 9: Appointment

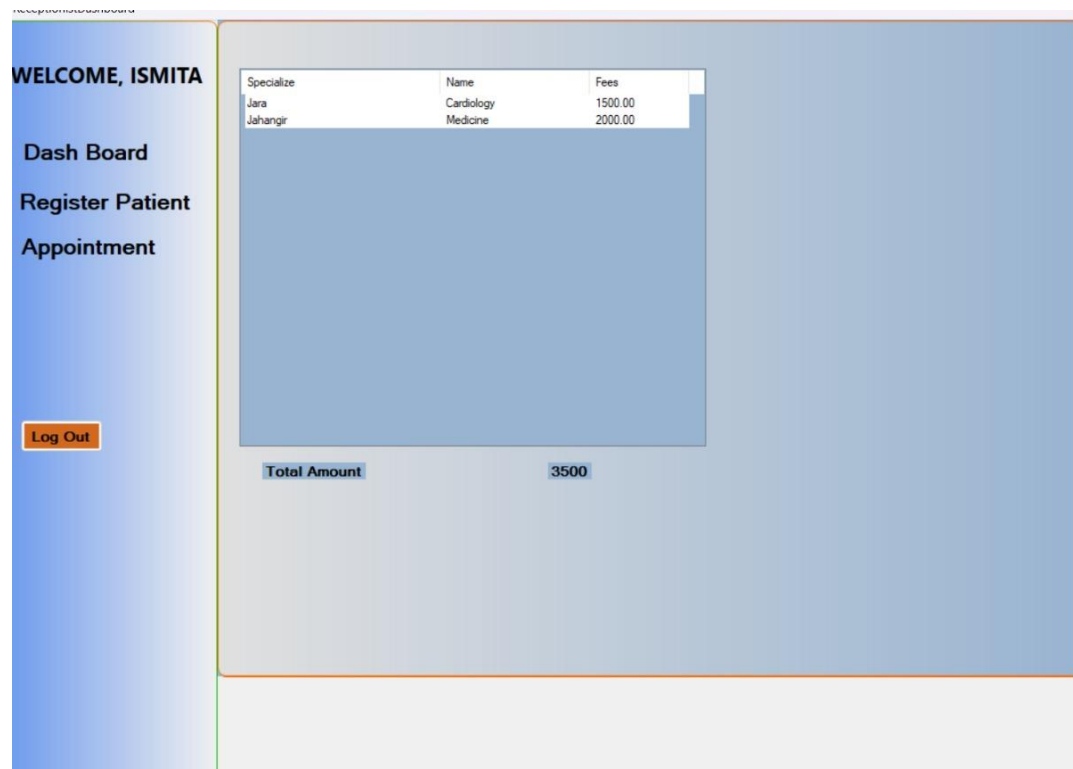


Figure 10: Billing